# 4

| | | | |
|---|---|---|---|
| | *5-4* | | *192.168.1.50* |
| | | | |
| | | PC1, | |
| | | PC2, | |
| | | PC3, 2 | |
| | | PC4, 1 | |
| | | PC5, 1 | |

# 4 OSPF

## 4.1

1 OSPF OSPF

2 OSPF

## 4.2

1 静态路由和 **OSPF** 协议 ping

2 OSPF OSPF 分析 **OSPF** 报文协议交互过程

## 4.3

（略）

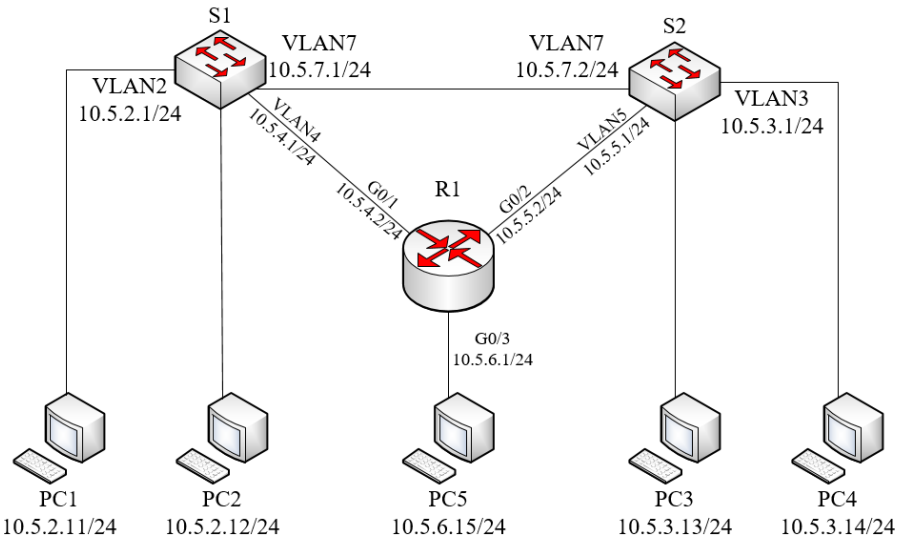
## 4.4

1 S5560 2 (S1 S2) MSR2600 1 (R1)

2 4/5

## 4.5

1




1.1 PC1 PC2 S1

1.2 R1 S1

1.3 R1 ping PC1 PC2 ping

R1 ping PC1 PC2

1.4 R1 S1

```
[MSR2600-1]display ip routing-table

Destinations : 16          Routes : 16

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0   0         127.0.0.1       InLoop0
10.5.4.0/24         Direct  0   0         10.5.4.2        GE0/1
10.5.4.2/32         Direct  0   0         127.0.0.1       InLoop0
10.5.4.255/32       Direct  0   0         10.5.4.2        GE0/1
10.5.5.0/24         Direct  0   0         10.5.5.2        GE0/2
10.5.5.2/32         Direct  0   0         127.0.0.1       InLoop0
10.5.5.255/32       Direct  0   0         10.5.5.2        GE0/2
```

```
10.5.6.0/24          Direct  0    0           10.5.6.1        GE0/3
10.5.6.1/32          Direct  0    0           127.0.0.1       InLoop0
10.5.6.255/32        Direct  0    0           10.5.6.1        GE0/3
127.0.0.0/8          Direct  0    0           127.0.0.1       InLoop0
127.0.0.1/32         Direct  0    0           127.0.0.1       InLoop0
127.255.255.255/32 Direct  0    0           127.0.0.1       InLoop0
224.0.0.0/4          Direct  0    0           0.0.0.0         NULL0
224.0.0.0/24         Direct  0    0           0.0.0.0         NULL0
255.255.255.255/32 Direct  0    0           127.0.0.1       InLoop0


<S1>display ip routing-table

Destinations : 24        Routes : 24

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0    0           127.0.0.1       InLoop0
10.5.2.0/24         Direct  0    0           10.5.2.1        Vlan2
10.5.2.0/32         Direct  0    0           10.5.2.1        Vlan2
10.5.2.1/32         Direct  0    0           127.0.0.1       InLoop0
10.5.2.255/32       Direct  0    0           10.5.2.1        Vlan2
10.5.4.0/24         Direct  0    0           10.5.4.1        Vlan4
10.5.4.0/32         Direct  0    0           10.5.4.1        Vlan4
10.5.4.1/32         Direct  0    0           127.0.0.1       InLoop0
10.5.4.255/32       Direct  0    0           10.5.4.1        Vlan4
10.5.7.0/24         Direct  0    0           10.5.7.1        Vlan7
10.5.7.0/32         Direct  0    0           10.5.7.1        Vlan7
10.5.7.1/32         Direct  0    0           127.0.0.1       InLoop0
10.5.7.255/32       Direct  0    0           10.5.7.1        Vlan7
127.0.0.0/8         Direct  0    0           127.0.0.1       InLoop0
127.0.0.0/32        Direct  0    0           127.0.0.1       InLoop0
127.0.0.1/32        Direct  0    0           127.0.0.1       InLoop0
127.255.255.255/32 Direct  0    0           127.0.0.1       InLoop0
169.254.0.0/16      Direct  0    0           169.254.123.51  Vlan1
169.254.0.0/32      Direct  0    0           169.254.123.51  Vlan1
169.254.123.51/32   Direct  0    0           127.0.0.1       InLoop0
169.254.255.255/32 Direct  0    0           169.254.123.51  Vlan1
224.0.0.0/4         Direct  0    0           0.0.0.0         NULL0
224.0.0.0/24        Direct  0    0           0.0.0.0         NULL0
255.255.255.255/32 Direct  0    0           127.0.0.1       InLoop0
```

1.5

PC1 PC2 S1 VLAN2 10.5.2.0/24 R1 S1 10.5.4.0/24 R1 10.5.4.0/24 10.5.5.0/24 10.5.6.0/24 10.5.2.0/24 R1 PC1 PC2 S1 10.5.4.0/24

2 R1

2.1 R1 ping PC ping

R1 ping PC1 PC2 PC5 PC3 PC4 ping

2.2 R1

```
[MSR2600-1]display ip routing-table

Destinations : 17        Routes : 17

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0    0           127.0.0.1       InLoop0
10.5.2.0/24         Static  60   0           10.5.4.1        GE0/1
10.5.4.0/24         Direct  0    0           10.5.4.2        GE0/1
10.5.4.2/32         Direct  0    0           127.0.0.1       InLoop0
10.5.4.255/32       Direct  0    0           10.5.4.2        GE0/1
10.5.5.0/24         Direct  0    0           10.5.5.2        GE0/2
10.5.5.2/32         Direct  0    0           127.0.0.1       InLoop0
10.5.5.255/32       Direct  0    0           10.5.5.2        GE0/2
```

```
10.5.6.0/24          Direct  0    0          10.5.6.1        GE0/3
10.5.6.1/32          Direct  0    0          127.0.0.1       InLoop0
10.5.6.255/32        Direct  0    0          10.5.6.1        GE0/3
127.0.0.0/8          Direct  0    0          127.0.0.1       InLoop0
127.0.0.1/32         Direct  0    0          127.0.0.1       InLoop0
127.255.255.255/32 Direct  0    0          127.0.0.1       InLoop0
224.0.0.0/4          Direct  0    0          0.0.0.0         NULL0
224.0.0.0/24         Direct  0    0          0.0.0.0         NULL0
255.255.255.255/32 Direct  0    0          127.0.0.1       InLoop0
```

2.3

R1 10.5.2.0/24 S1 PC1 PC2 PC5 10.5.6.0/24 PC3 PC4 10.5.3.0/24

**3** 2

**4** S1 R1 OSPF

4.1 R1 ping PC ping

```
[MSR2600-1]ping 10.5.2.11
PING 10.5.2.11: 56 data bytes, press CTRL_C to break
Reply from 10.5.2.11: bytes=56 Sequence=1 ttl=127 time=2 ms
Reply from 10.5.2.11: bytes=56 Sequence=2 ttl=127 time=1 ms
Reply from 10.5.2.11: bytes=56 Sequence=3 ttl=127 time=1 ms
Reply from 10.5.2.11: bytes=56 Sequence=4 ttl=127 time=2 ms
Reply from 10.5.2.11: bytes=56 Sequence=5 ttl=127 time=1 ms

--- 10.5.2.11 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms


[MSR2600-1]ping 10.5.2.12
PING 10.5.2.12: 56  data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.5.2.12 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss


[MSR2600-1]ping 10.5.3.13
PING 10.5.3.13: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.5.3.13 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss


[MSR2600-1]ping 10.5.3.14
PING 10.5.3.14: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 10.5.3.14 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss


[MSR2600-1]ping 10.5.6.15
PING 10.5.6.15: 56 data bytes, press CTRL_C to break
Reply from 10.5.6.15: bytes=56 Sequence=1 ttl=128 time=1 ms
Reply from 10.5.6.15: bytes=56 Sequence=2 ttl=128 time=1 ms
Reply from 10.5.6.15: bytes=56 Sequence=3 ttl=128 time=1 ms
Reply from 10.5.6.15: bytes=56 Sequence=4 ttl=128 time=1 ms
Reply from 10.5.6.15: bytes=56 Sequence=5 ttl=128 time=1 ms

--- 10.5.6.15 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/1 ms
```

4.2

| | Destination/Mask | Protocol | Pref | Cost | Nexthop | Interface |
|---|---|---|---|---|---|---|
| S1 | 10.5.2.0/24 | Direct | 0 | 0 | 10.5.2.1 | Vlan2 |
| | 10.5.4.0/24 | Direct | 0 | 0 | 10.5.4.1 | Vlan4 |
| | 10.5.5.0/24 | O_INTRA | 10 | 2 | 10.5.4.2 | Vlan4 |
| | 10.5.6.0/24 | O_INTRA | 10 | 2 | 10.5.4.2 | Vlan4 |
| | 10.5.7.0/24 | Direct | 0 | 0 | 10.5.7.1 | Vlan7 |
| R1 | 10.5.2.0/24 | O_INTRA | 10 | 2 | 10.5.4.1 | GE0/1 |
| | 10.5.4.0/24 | Direct | 0 | 0 | 10.5.4.2 | GE0/1 |
| | 10.5.5.0/24 | Direct | 0 | 0 | 10.5.5.2 | GE0/2 |
| | 10.5.6.0/24 | Direct | 0 | 0 | 10.5.6.1 | GE0/3 |
| | 10.5.7.0/24 | O_INTRA | 10 | 2 | 10.5.4.1 | GE0/1 |

4.3 PC /
R1 OSPF 10.5.2.0/24 10.5.7.0/24 S1
10.5.6.0/24 R1 PC5 S2 OSPF R1 10.5.3.0/24
PC3 PC4 PC2 PC2
OSPF

4.4 display OSPF

```
[MSR2600-1]display ospf 1    peer verbose

                OSPF Process 1 with Router ID 1.1.1.1
                        Neighbors

 Area 0.0.0.1 interface 10.5.4.2(GigabitEthernet0/1)'s neighbors
 Router ID: 10.5.4.1             Address: 10.5.4.1             GR State: Normal
    State: Full    Mode: Nbr is master    Priority: 1
    DR: 10.5.4.1    BDR: 10.5.4.2    MTU: 0
```

```
    Options is 0x42 (-|O|-|-|-|E|-)
    Dead timer due in 37    sec
    Neighbor is up for 00:19:03
    Authentication Sequence: [ 0 ]
    Neighbor state change count: 6
    BFD status: Disabled

[MSR2600-1]display ospf 1    peer

            OSPF Process 1 with Router ID 1.1.1.1
                    Neighbor Brief Information

 Area: 0.0.0.1
 Router ID          Address          Pri Dead-Time   State             Interface
 10.5.4.1           10.5.4.1         1   35          Full/DR           GE0/1

[MSR2600-1]display ospf 1 verbose

            OSPF Process 1 with Router ID 1.1.1.1
                    OSPF Protocol Information

 RouterID: 1.1.1.1                Router type:
 Route tag: 0
 Multi-VPN-Instance is not enabled
 Ext-community type: Domain ID 0x5, Route Type 0x306, Router ID 0x107
 Domain ID: 0.0.0.0
 Opaque capable
 Isolation: Disabled
 ISPF is enabled
 SPF-schedule-interval: 5 50 200
 LSA generation interval: 5 50 200
 LSA arrival interval: 1000
 Transmit pacing: Interval: 20 Count: 3
 Default ASE parameters: Metric: 1 Tag: 1 Type: 2
 Route preference: 10
 ASE route preference: 150
 SPF calculation count: 17
 RFC 1583 compatible
 Fast-reroute: Remote-lfa Disabled
    Maximum-cost: 4294967295
 Node-Protecing Preference: 40
 Lowest-cost Preference: 20
 Graceful restart interval: 120
 SNMP trap rate limit interval: 10    Count: 7
 Area count: 1      NSSA area count: 0
 Normal areas with up interfaces: 1
 NSSA areas with up interfaces: 0
 Up interfaces: 3
 ExChange/Loading neighbors: 0
 Full neighbors: 1
 Calculation trigger type: N/A
 Current calculation type: N/A
 Current calculation phase: N/A
 Process reset state: N/A
 Current reset type: N/A
 Next reset type: N/A
 Reset prepare message replied: -/-/-/-
 Reset process message replied: -/-/-/-
 Reset phase of module:
    M-N/A, P-N/A, L-N/A, C-N/A, R-N/A
 MPLS segment routing: Disabled
    Segment routing adjacency      : Disabled
    Effective SRGB                 : 16000    24000
    Segment routing local block : 15000    15999
    Segment routing tunnel count: 0

 Area: 0.0.0.1                (MPLS TE not enabled)
 Authentication type: None        Area flag: Normal
 SPF scheduled count: 3
 ExChange/Loading neighbors: 0
 Up interfaces: 3

 Interface: 10.5.4.2 (GigabitEthernet0/1)
 Cost: 1            State: BDR            Type: Broadcast        MTU: 1500
 Priority: 1
```

```
  Designated router: 10.5.4.1
  Backup designated router: 10.5.4.2
  Timers: Hello 10, Dead 40, Poll 40, Retransmit 5, Transmit Delay 1
  FRR backup: Enabled
  FRR TI-LFA: Enabled
  FRR remote-lfa: Enabled
  Enabled by network configuration

  Interface: 10.5.5.2 (GigabitEthernet0/2)
  Cost: 1          State: DR          Type: Broadcast       MTU: 1500
  Priority: 1
  Designated router: 10.5.5.2
  Backup designated router: 0.0.0.0
  Timers: Hello 10, Dead 40, Poll 40, Retransmit 5, Transmit Delay 1
  FRR backup: Enabled
  FRR TI-LFA: Enabled
  FRR remote-lfa: Enabled
  Enabled by network configuration

  Interface: 10.5.6.1 (GigabitEthernet0/3)
  Cost: 1          State: DR          Type: Broadcast       MTU: 1500
  Priority: 1
  Designated router: 10.5.6.1
  Backup designated router: 0.0.0.0
  Timers: Hello 10, Dead 40, Poll 40, Retransmit 5, Transmit Delay 1
  FRR backup: Enabled
  FRR TI-LFA: Enabled
  FRR remote-lfa: Enabled
  Enabled by network configuration

MSR2600-1]display ip routing-table

Destinations : 18          Routes : 18

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0   0         127.0.0.1       InLoop0
10.5.2.0/24         O_INTRA 10  2           10.5.4.1        GE0/1
10.5.4.0/24         Direct  0   0         10.5.4.2        GE0/1
10.5.4.2/32         Direct  0   0         127.0.0.1       InLoop0
10.5.4.255/32       Direct  0   0         10.5.4.2        GE0/1
10.5.5.0/24         Direct  0   0         10.5.5.2        GE0/2
10.5.5.2/32         Direct  0   0         127.0.0.1       InLoop0
10.5.5.255/32       Direct  0   0         10.5.5.2        GE0/2
10.5.6.0/24         Direct  0   0         10.5.6.1        GE0/3
10.5.6.1/32         Direct  0   0         127.0.0.1       InLoop0
10.5.6.255/32       Direct  0   0         10.5.6.1        GE0/3
10.5.7.0/24         O_INTRA 10  2           10.5.4.1        GE0/1
127.0.0.0/8         Direct  0   0         127.0.0.1       InLoop0
127.0.0.1/32        Direct  0   0         127.0.0.1       InLoop0
127.255.255.255/32  Direct  0   0         127.0.0.1       InLoop0
224.0.0.0/4         Direct  0   0         0.0.0.0         NULL0
224.0.0.0/24        Direct  0   0         0.0.0.0         NULL0
255.255.255.255/32  Direct  0   0         127.0.0.1       InLoop0


[MSR2600-1]display ospf routing

          OSPF Process 1 with Router ID 1.1.1.1
                  Routing Table

  Routing for network
  Destination     Cost    Type    NextHop         AdvRouter       Area
  10.5.5.0/24     1       Stub    0.0.0.0         1.1.1.1         0.0.0.1
  10.5.2.0/24     2       Stub    10.5.4.1        10.5.4.1        0.0.0.1
  10.5.6.0/24     1       Stub    0.0.0.0         1.1.1.1         0.0.0.1
  10.5.7.0/24     2       Stub    10.5.4.1        10.5.4.1        0.0.0.1
  10.5.4.0/24     1       Transit 0.0.0.0         10.5.4.1        0.0.0.1

  Total nets: 5
  Intra area: 5  Inter area: 0  ASE: 0  NSSA: 0

[MSR2600-1]display ospf interface GigabitEthernet 0/1


          OSPF Process 1 with Router ID 1.1.1.1
```

```
                Interfaces

 Area: 0.0.0.1

 Interface: 10.5.4.2 (GigabitEthernet0/1)
 Cost: 1          State: BDR          Type: Broadcast       MTU: 1500
 Priority: 1
 Designated router: 10.5.4.1
 Backup designated router: 10.5.4.2
 Timers: Hello 10, Dead 40, Poll 40, Retransmit 5, Transmit Delay 1
 FRR backup: Enabled
 FRR TI-LFA: Enabled
 FRR remote-lfa: Enabled
 Enabled by network configuration
```

4.5 OSPF

R1 OSPF 1 Router ID 1.1.1.1 area 0.0.0.1 S1 1 Full LSDB GE0/1 Broadcast 10.5.4.0/24 S1 DR R1 BDR Hello/Dead 10s 40s OSPF 10.5.2.0/24 10.5.7.0/24 Cost 2 R1 SPF

5 S2 S2 OSPF

5.1 PC

PC1 PC4 PC5 PC1 PC4 S1 S2 PC2 PC3

5.2 PC1 tracert PC1-PC4

```
C:\Users\Administrator>tracert -d  10.5.3.14

     30          10.5.3.14

  1    <1        <1        <1      10.5.2.1
  2    <1        <1        <1      10.5.7.2
  3    <1        <1        <1      10.5.3.14
```

5.3 S1

```
[S1]dis ip routing-table

Destinations : 27        Routes : 28

Destination/Mask    Proto   Pre  Cost      NextHop         Interface
0.0.0.0/32          Direct  0    0         127.0.0.1       InLoop0
10.5.2.0/24         Direct  0    0         10.5.2.1        Vlan2
10.5.2.0/32         Direct  0    0         10.5.2.1        Vlan2
10.5.2.1/32         Direct  0    0         127.0.0.1       InLoop0
10.5.2.255/32       Direct  0    0         10.5.2.1        Vlan2
10.5.3.0/24         O_INTRA 10   2           10.5.7.2        Vlan7
10.5.4.0/24         Direct  0    0         10.5.4.1        Vlan4
10.5.4.0/32         Direct  0    0         10.5.4.1        Vlan4
10.5.4.1/32         Direct  0    0         127.0.0.1       InLoop0
10.5.4.255/32       Direct  0    0         10.5.4.1        Vlan4
10.5.5.0/24         O_INTRA 10   2           10.5.4.2        Vlan4
                                             10.5.7.2        Vlan7
```

```
10.5.6.0/24          O_INTRA 10    2           10.5.4.2        Vlan4
10.5.7.0/24          Direct  0     0           10.5.7.1        Vlan7
10.5.7.0/32          Direct  0     0           10.5.7.1        Vlan7
10.5.7.1/32          Direct  0     0           127.0.0.1       InLoop0
10.5.7.255/32        Direct  0     0           10.5.7.1        Vlan7
127.0.0.0/8          Direct  0     0           127.0.0.1       InLoop0
127.0.0.0/32         Direct  0     0           127.0.0.1       InLoop0
127.0.0.1/32         Direct  0     0           127.0.0.1       InLoop0
127.255.255.255/32 Direct    0     0           127.0.0.1       InLoop0
169.254.0.0/16       Direct  0     0           169.254.123.51  Vlan1
169.254.0.0/32       Direct  0     0           169.254.123.51  Vlan1
169.254.123.51/32    Direct  0     0           127.0.0.1       InLoop0
169.254.255.255/32 Direct    0     0           169.254.123.51  Vlan1
224.0.0.0/4          Direct  0     0           0.0.0.0         NULL0
224.0.0.0/24         Direct  0     0           0.0.0.0         NULL0
255.255.255.255/32 Direct    0     0           127.0.0.1       InLoop0
```

6 S1 S2 PC1 PC4

6.1 PC1 PC4

S1 S2 PC1 PC4

6.2 tracert

```
C:\Users\Administrator>tracert -d  10.5.3.14

      30            10.5.3.14

  1    <1        <1        <1       10.5.2.1
  2    <1        <1        <1       10.5.4.2
  3     1  ms     25  ms     <1      10.5.5.1
  4    <1        <1        <1       10.5.3.14
```

6.3 S1

```
[S1]dis ip routing-table

Destinations : 23          Routes : 23

Destination/Mask    Proto   Pre  Cost      NextHop         Interface
0.0.0.0/32          Direct  0    0         127.0.0.1       InLoop0
10.5.2.0/24         Direct  0    0         10.5.2.1        Vlan2
10.5.2.0/32         Direct  0    0         10.5.2.1        Vlan2
10.5.2.1/32         Direct  0    0         127.0.0.1       InLoop0
10.5.2.255/32       Direct  0    0         10.5.2.1        Vlan2
10.5.3.0/24         O_INTRA 10   3         10.5.4.2        Vlan4
10.5.4.0/24         Direct  0    0         10.5.4.1        Vlan4
10.5.4.0/32         Direct  0    0         10.5.4.1        Vlan4
10.5.4.1/32         Direct  0    0         127.0.0.1       InLoop0
10.5.4.255/32       Direct  0    0         10.5.4.1        Vlan4
10.5.5.0/24         O_INTRA 10   2         10.5.4.2        Vlan4
10.5.6.0/24         O_INTRA 10   2         10.5.4.2        Vlan4
127.0.0.0/8         Direct  0    0         127.0.0.1       InLoop0
127.0.0.0/32        Direct  0    0         127.0.0.1       InLoop0
127.0.0.1/32        Direct  0    0         127.0.0.1       InLoop0
127.255.255.255/32 Direct   0    0         127.0.0.1       InLoop0
169.254.0.0/16      Direct  0    0         169.254.123.51  Vlan1
169.254.0.0/32      Direct  0    0         169.254.123.51  Vlan1
169.254.123.51/32   Direct  0    0         127.0.0.1       InLoop0
169.254.255.255/32 Direct   0    0         169.254.123.51  Vlan1
224.0.0.0/4         Direct  0    0         0.0.0.0         NULL0
224.0.0.0/24        Direct  0    0         0.0.0.0         NULL0
255.255.255.255/32 Direct   0    0         127.0.0.1       InLoop0
```

6.4 S1

5 S1 10.5.7.0/24 S1—S2 10.5.3.0/24

10.5.7.2 Cost 2 10.5.4.2 Cost 3 S1→R1→S2 10.5.5.0/24

10.5.7.2 R1 OSPF LSA SPF

7 OSPF

7.1 PC2 PC3 OSPF OSPF

OSPF Hello Router ID Area ID Hello/Dead DR/BDR Hello Down/Init 2-Way DR/BDR ExStart/Exchange DD LSDB LSA LSR LSU LSA LSAck Full SPF R1 10.5.2.0/24 10.5.7.0/24 S1 10.5.5.0/24 10.5.6.0/24 S2 S1 10.5.3.0/24 10.5.5.0/24

7.2 S1 S2 OSPF

S1 S2 10.5.7.0/24 Hello down Dead Timer Full Down Router-LSA LSU/LSAck R1 LSDB SPF S1 10.5.3.0/24 R1 Cost 2 3 tracert 10.5.2.1→10.5.4.2→10.5.5.1→10.5.3.14 Hello 2-Way Exchange Loading Full LSDB 10.5.7.2

## 4.6

8

8.1

0.0.0.0/0

up IP

OSPF RIP

8.2 OSPF

OSPF Hello/Dead

LSDB

SPF

8.3 OSPF

OSPF O_INTRA IP

Pref Cost Direct

10.5.2.0/24 10.5.3.0/24 OSPF OSPF

## 4.7

9 OSPF +

1

```
[R1]display ospf peer

        OSPF Process 1 with Router ID 1.1.1.1
              Neighbor Brief Information

 Area: 0.0.0.1
 Router ID       Address         Pri Dead-Time  State             Interface
 10.6.4.2        10.6.4.1        1   39         Full/BDR          GE0/1
 10.6.5.1        10.6.5.1        1   36         Full/DR           GE0/2
[R1]
[R1]display ospf interface


        OSPF Process 1 with Router ID 1.1.1.1
                Interfaces

 Area: 0.0.0.1
 IP Address      Type      State    Cost  Pri   DR              BDR
 10.6.4.2        Broadcast DR       1     1     10.6.4.2        10.6.4.1
 10.6.5.2        Broadcast BDR      1     1     10.6.5.1        10.6.5.2
 10.6.6.1        Broadcast DR       1     1     10.6.6.1        0.0.0.0
[R1]
[R1]display ospf routing

        OSPF Process 1 with Router ID 1.1.1.1
                 Routing Table

 Routing for network
 Destination        Cost     Type    NextHop         AdvRouter       Area
 10.6.4.0/24        1        Transit 0.0.0.0         1.1.1.1         0.0.0.1
 10.6.3.0/24        2        Stub    10.6.5.1        10.6.5.1        0.0.0.1
 10.6.7.0/24        2        Transit 10.6.4.1        10.6.4.2        0.0.0.1
 10.6.7.0/24        2        Transit 10.6.5.1        10.6.4.2        0.0.0.1
 10.6.6.0/24        1        Stub    0.0.0.0         1.1.1.1         0.0.0.1
 10.6.2.0/24        2        Stub    10.6.4.1        10.6.4.2        0.0.0.1
 10.6.5.0/24        1        Transit 0.0.0.0         10.6.5.1        0.0.0.1

 Total nets: 7
 Intra area: 7  Inter area: 0  ASE: 0  NSSA: 0
```

display ospf peer R1 Router ID 1.1.1.1 Area 0.0.0.1 GE0/1 GE0/2 10.6.4.1 10.6.5.1 Full GE0/1 Full/BDR GE0/2 Full/DR R1 Hello

display ospf interface R1 OSPF 10.6.4.2 10.6.5.2 10.6.6.1 Broadcast Cost 1 10.6.4.2 R1 DR 10.6.5.2 R1 BDR

10.6.6.1 R1 DR BDR=0.0.0.0 OSPF

display ospf routing R1 7 Intra-area Inter-area

R1 OSPF 10.6.2.0/24 10.6.3.0/24 10.6.4.1 10.6.5.1

10.6.7.0/24 Cost 10.6.4.1 10.6.5.1 OSPF

2

```
[R1]display ospf lsdb

         OSPF Process 1 with Router ID 1.1.1.1
                 Link State Database

                         Area: 0.0.0.1
 Type      LinkState ID    AdvRouter       Age  Len   Sequence  Metric
 Router    10.6.4.2        10.6.4.2        1558 60    8000000B  0
 Router    10.6.5.1        10.6.5.1        238  60    8000002B  0
 Router    1.1.1.1         1.1.1.1         237  60    80000026  0
 Network   10.6.7.1        10.6.4.2        1549 32    80000002  0
 Network   10.6.4.2        1.1.1.1         4    32    80000003  0
 Network   10.6.5.1        10.6.5.1        238  32    80000001  0
```

display ospf lsdb Area 0.0.0.1 Router-LSA Network-LSA

3 Router-LSA 3 Network-LSA

LSDB

Router-LSA Network-LSA

Broadcast DR Type 2 Network-LSA

LSA Age Sequence OSPF

3

```
[R1]display ospf event peer

        OSPF Process 1 with Router ID 1.1.1.1
                Neighbors Log

Date       Time     Local Address   Remote Address  Router ID       Reason
2011-01-01 00:41:52 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
2011-01-01 00:33:53 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
2011-01-01 00:33:36 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
2011-01-01 00:33:06 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
2011-01-01 00:29:27 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
2011-01-01 00:29:12 10.6.5.2        10.6.5.1        10.6.5.1        IntPhyChange
[R1]
[R1]display ospf event lsa-f
[R1]display ospf event lsa-flush

         OSPF Process 1 with Router ID 1.1.1.1
                 LSA Flush Log

 Date: 2011-01-01 Time: 00:41:52 Received MaxAge LSA from 10.6.4.1
 Type: 2   LS ID: 10.6.5.1        AdvRtr: 10.6.5.1         Seq#: 80000001

 Date: 2011-01-01 Time: 00:29:12 Received MaxAge LSA from 10.6.4.1
 Type: 2   LS ID: 10.6.5.1        AdvRtr: 10.6.5.1         Seq#: 80000001
```

display ospf event peer R1 10.6.5.2 10.6.5.1 IntPhyChange OSPF up/down

display ospf event lsa-flush R1 10.6.4.1 MaxAge LSA Type 2 LS ID 10.6.5.1 LSA OSPF LSA

OSPF LSDB

4

```
[R1]display ospf event spf

        OSPF Process 1 with Router ID 1.1.1.1
                SPF Log

Date       Time     Duration   Intra    Inter    External Reason
2011-01-01 00:42:42 2562.846   3        0        0        Intra-area LSA
2011-01-01 00:42:01 2521.755   1        0        0        Intra-area LSA
2011-01-01 00:41:59 2519.456   2        0        0        Intra-area LSA
2011-01-01 00:41:58 2518.856   1        0        0        Intra-area LSA
2011-01-01 00:41:53 2513.465   1        0        0        Intra-area LSA
2011-01-01 00:41:52 2512.86    4        0        0        Intra-area LSA
2011-01-01 00:36:30 2190.186   3        0        0        Intra-area LSA
2011-01-01 00:35:46 2146.456   2        0        0        Intra-area LSA
2011-01-01 00:35:46 2146.185   1        0        0        Intra-area LSA
2011-01-01 00:33:53 2033.456   1        0        0        Intra-area LSA
2011-01-01 00:33:52 2032.565   0        0        0        Intra-area LSA
2011-01-01 00:33:38 2018.456   1        0        0        Intra-area LSA
2011-01-01 00:33:36 2016.455   1        0        0        Intra-area LSA
2011-01-01 00:33:35 2015.435   0        0        0        Intra-area LSA
2011-01-01 00:33:09 1989.456   2        0        0        Intra-area LSA
2011-01-01 00:33:08 1988.456   1        0        0        Intra-area LSA
2011-01-01 00:33:06 1986.455   1        0        0        Intra-area LSA
2011-01-01 00:33:05 1985.455   0        0        0        Intra-area LSA
2011-01-01 00:32:55 1975.985   1        0        0        Intra-area LSA
2011-01-01 00:32:49 1969.705   1        0        0        Intra-area LSA
2011-01-01 00:32:44 1964.665   1        0        0        Intra-area LSA
2011-01-01 00:32:44 1964.456   2        0        0        Intra-area LSA
2011-01-01 00:32:43 1963.455   1        0        0        Intra-area LSA
2011-01-01 00:32:41 1961.455   1        0        0        Intra-area LSA
2011-01-01 00:32:41 1961.105   1        0        0        Intra-area LSA
2011-01-01 00:32:39 1959.456   1        0        0        Intra-area LSA
2011-01-01 00:30:05 1805.455   1        0        0        Intra-area LSA
2011-01-01 00:30:04 1804.435   0        0        0        Intra-area LSA
2011-01-01 00:30:03 1803.456   2        0        0        Intra-area LSA
2011-01-01 00:30:03 1803.25    1        0        0        Intra-area LSA
2011-01-01 00:30:01 1801.455   1        0        0        Intra-area LSA
2011-01-01 00:29:59 1799.985   0        0        0        Intra-area LSA
2011-01-01 00:29:51 1791.456   2        0        0        Intra-area LSA
2011-01-01 00:29:50 1790.985   1        0        0        Intra-area LSA
2011-01-01 00:29:27 1767.456   1        0        0        Intra-area LSA
2011-01-01 00:29:26 1766.965   0        0        0        Intra-area LSA
2011-01-01 00:29:15 1755.456   2        0        0        Intra-area LSA
2011-01-01 00:29:14 1754.765   1        0        0        Intra-area LSA
2011-01-01 00:29:12 1752.116   4        0        0        Intra-area LSA
2011-01-01 00:29:01 1741.46    3        0        0        Intra-area LSA
2011-01-01 00:28:16 1696.456   2        0        0        Intra-area LSA
2011-01-01 00:28:16 1696.55    1        0        0        Intra-area LSA
2011-01-01 00:23:18 1398.115   1        0        0        Intra-area LSA
2011-01-01 00:20:50 1250.15    0        0        0        Intra-area LSA
2011-01-01 00:20:49 1249.55    0        0        0        Intra-area LSA
2011-01-01 00:20:42 1242.795   0        0        0        Intra-area LSA
2011-01-01 00:18:18 1098.105   1        0        0        Intra-area LSA
2011-01-01 00:18:10 1090.66    1        0        0        Intra-area LSA
2011-01-01 00:18:02 1082.685   0        0        0        Intra-area LSA
2011-01-01 00:18:02 1082.433   0        0        0        Intra-area LSA
2011-01-01 00:15:12 912.455    1        0        0        Intra-area LSA
2011-01-01 00:15:10 910.446    1        0        0        Intra-area LSA
2011-01-01 00:06:17 377.456    1        0        0        Intra-area LSA
2011-01-01 00:06:08 368.676    1        0        0        Intra-area LSA
2011-01-01 00:06:05 365.746    1        0        0        Intra-area LSA
2011-01-01 00:06:01 361.699    1        0        0        Intra-area LSA
```

display ospf event spf R1 SPF Intra-area LSA Inter External 0 OSPF SPF LSA OSPF SPF LSDB OSPF SPF

5

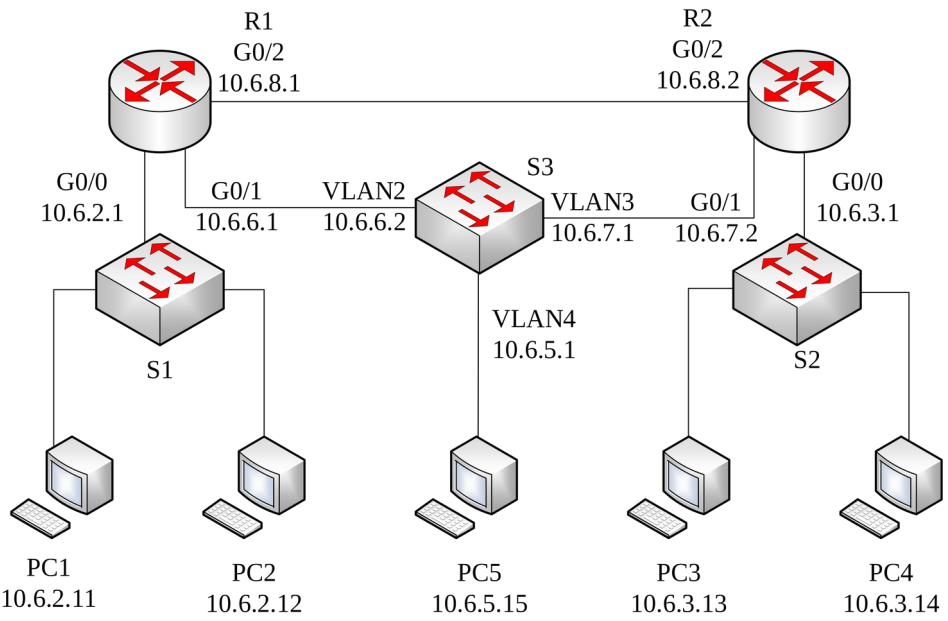
```
[R1]display ospf statistics

        OSPF Process 1 with Router ID 1.1.1.1
                Statistics

 I/O statistics
  Type                    Input      Output
  Hello                   281        637
  DB Description          11         14
  Link-State Req          1          1
  Link-State Update       54         39
  Link-State Ack          39         50

 LSAs originated by this router
  Router  : 38
  Network : 2
  Sum-Net : 0
  Sum-Asbr: 0
  External: 0
  NSSA    : 0
  Opq-Link: 0
  Opq-Area: 0
  Opq-AS  : 0

 LSAs originated: 40  LSAs received: 64

 Routing table:
   Intra area: 7  Inter area: 0  ASE/NSSA: 0
```

display ospf statistics Hello 281 637 Hello OSPF

DB Description Link-State Request Link-State Update Link-State Ack Exchange Loading DD LSA LSR LSU LSAck

LSA R1 40 LSA 64 LSA Router-LSA 38 Network-LSA 2 R1 7 Inter-area ASE/NSSA 0 OSPF

10 RIP +




## 10.1 RIP

1 IP VLAN 2 VLAN 3 VLAN 4 VLAN VLAN

2 RIPv2

3 RIP

4 display ip routing-table RIP

5 1 2

6 1 2

R1

```
[RA] interface GigabitEthernet G/0/0
[RA-GigabitEthernetG/0/0] ip address 10.6.2.1 255.255.255.0
[RA-GigabitEthernetG/0/0] quit

[RA] interface GigabitEthernet G/0/1
[RA-GigabitEthernetG/0/1] ip address 10.6.6.1 255.255.255.0
[RA-GigabitEthernetG/0/1] quit

[RA] interface GigabitEthernet G/0/2
[RA-GigabitEthernetG/0/2] ip address 10.6.8.1 255.255.255.0
[RA-GigabitEthernetG/0/2] quit

[RA] rip 1
[RA-rip-1] version 2
```

```
[RA-rip-1] undo summary
[RA-rip-1] network 10.6.2.0
[RA-rip-1] network 10.6.6.0
[RA-rip-1] network 10.6.8.0
[RA-rip-1] quit
```

R2

```
[RB] interface GigabitEthernet G/0/0
[RB-GigabitEthernetG/0/0] ip address 10.6.3.1 255.255.255.0
[RB-GigabitEthernetG/0/0] quit

[RB] interface GigabitEthernet G/0/1
[RB-GigabitEthernetG/0/1] ip address 10.6.7.2 255.255.255.0
[RB-GigabitEthernetG/0/1] quit

[RB] interface GigabitEthernet G/0/2
[RB-GigabitEthernetG/0/2] ip address 10.6.8.2 255.255.255.0
[RB-GigabitEthernetG/0/2] quit

[RB] rip 1
[RB-rip-1] version 2
[RB-rip-1] undo summary
[RB-rip-1] network 10.6.3.0
[RB-rip-1] network 10.6.7.0
[RB-rip-1] network 10.6.8.0
[RB-rip-1] quit
```

S3

```
[SW] vlan 2
[SW-vlan2] quit
[SW] vlan 3
[SW-vlan3] quit
[SW] vlan 4
[SW-vlan4] quit

[SW] interface Vlan-interface 2
[SW-Vlan-interface2] ip address 10.6.6.2 255.255.255.0
[SW-Vlan-interface2] quit
```

```
[SW] interface Vlan-interface 3
[SW-Vlan-interface3] ip address 10.6.7.1 255.255.255.0
[SW-Vlan-interface3] quit

[SW] interface Vlan-interface 4
[SW-Vlan-interface4] ip address 10.6.5.1 255.255.255.0
[SW-Vlan-interface4] quit

[SW] rip 1
[SW-rip-1] version 2
[SW-rip-1] undo summary
[SW-rip-1] network 10.6.5.0
[SW-rip-1] network 10.6.6.0
[SW-rip-1] network 10.6.7.0
[SW-rip-1] quit
```

10.2

S3

```
<SW>display ip routing-table

Destinations : 23          Routes : 24

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0   0         127.0.0.1       InLoop0
10.6.2.0/24         RIP     100 1         10.6.6.1        Vlan2
10.6.3.0/24         RIP     100 1         10.6.7.2        Vlan3
10.6.5.0/24         Direct  0   0         10.6.5.1        Vlan4
10.6.5.0/32         Direct  0   0         10.6.5.1        Vlan4
10.6.5.1/32         Direct  0   0         127.0.0.1       InLoop0
10.6.5.255/32       Direct  0   0         10.6.5.1        Vlan4
10.6.6.0/24         Direct  0   0         10.6.6.2        Vlan2
10.6.6.0/32         Direct  0   0         10.6.6.2        Vlan2
10.6.6.2/32         Direct  0   0         127.0.0.1       InLoop0
10.6.6.255/32       Direct  0   0         10.6.6.2        Vlan2
10.6.7.0/24         Direct  0   0         10.6.7.1        Vlan3
10.6.7.0/32         Direct  0   0         10.6.7.1        Vlan3
```

| | | | | | |
|---|---|---|---|---|---|
| 10.6.7.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 10.6.7.255/32 | Direct | 0 | 0 | 10.6.7.1 | Vlan3 |
| 10.6.8.0/24 | RIP | 100 | 1 | 10.6.6.1 | Vlan2 |
| | | | | 10.6.7.2 | Vlan3 |
| 127.0.0.0/8 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 127.0.0.0/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 127.0.0.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 127.255.255.255/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 224.0.0.0/4 | Direct | 0 | 0 | 0.0.0.0 | NULL0 |
| 224.0.0.0/24 | Direct | 0 | 0 | 0.0.0.0 | NULL0 |
| 255.255.255.255/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |

R1

<RA>display ip routing-table

Destinations : 20 Routes : 21

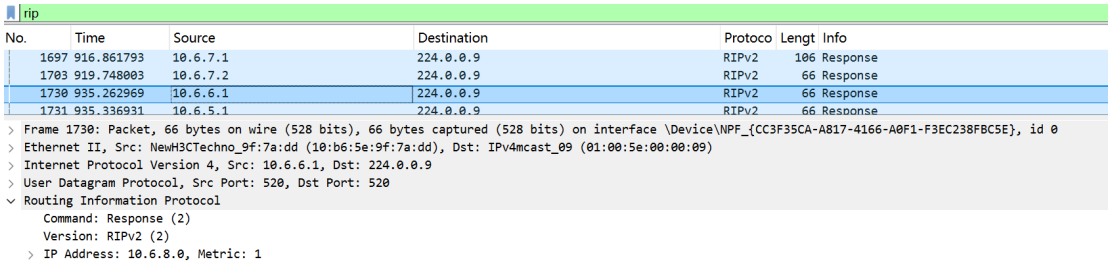
| Destination/Mask | Proto | Pre | Cost | NextHop | Interface |
|---|---|---|---|---|---|
| 0.0.0.0/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 10.6.2.0/24 | Direct | 0 | 0 | 10.6.2.1 | GE0/0 |
| 10.6.2.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 10.6.2.255/32 | Direct | 0 | 0 | 10.6.2.1 | GE0/0 |
| 10.6.3.0/24 | RIP | 100 | 1 | 10.6.8.2 | GE0/2 |
| 10.6.5.0/24 | RIP | 100 | 1 | 10.6.6.2 | GE0/1 |
| 10.6.6.0/24 | Direct | 0 | 0 | 10.6.6.1 | GE0/1 |
| 10.6.6.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 10.6.6.255/32 | Direct | 0 | 0 | 10.6.6.1 | GE0/1 |
| 10.6.7.0/24 | RIP | 100 | 1 | 10.6.6.2 | GE0/1 |
| | RIP | 100 | 1 | 10.6.8.2 | GE0/2 |
| 10.6.8.0/24 | Direct | 0 | 0 | 10.6.8.1 | GE0/2 |
| 10.6.8.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 10.6.8.255/32 | Direct | 0 | 0 | 10.6.8.1 | GE0/2 |
| 127.0.0.0/8 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 127.0.0.1/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 127.255.255.255/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |
| 224.0.0.0/4 | Direct | 0 | 0 | 0.0.0.0 | NULL0 |
| 224.0.0.0/24 | Direct | 0 | 0 | 0.0.0.0 | NULL0 |
| 255.255.255.255/32 | Direct | 0 | 0 | 127.0.0.1 | InLoop0 |

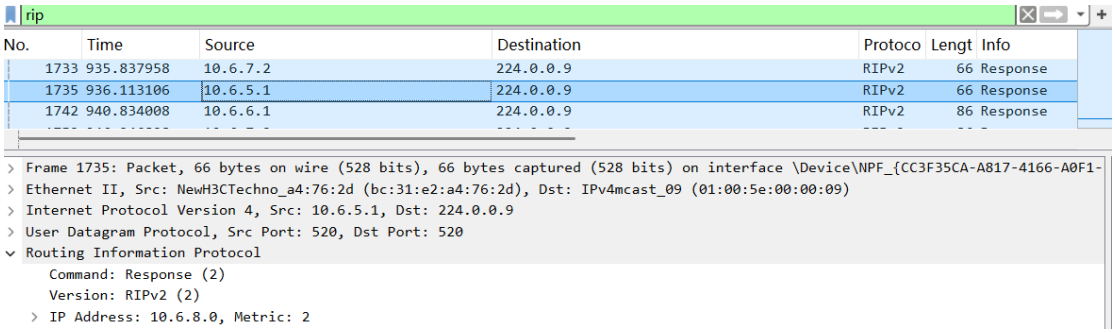
R2:

```
[RB]display ip routing-table

Destinations : 20          Routes : 21

Destination/Mask    Proto   Pre Cost      NextHop         Interface
0.0.0.0/32          Direct  0   0         127.0.0.1       InLoop0
10.6.2.0/24         RIP     100 1         10.6.8.1        GE0/2
10.6.3.0/24         Direct  0   0         10.6.3.1        GE0/0
10.6.3.1/32         Direct  0   0         127.0.0.1       InLoop0
10.6.3.255/32       Direct  0   0         10.6.3.1        GE0/0
10.6.5.0/24         RIP     100 1         10.6.7.1        GE0/1
10.6.6.0/24         RIP     100 1         10.6.7.1        GE0/1
                    RIP     100 1         10.6.8.1        GE0/2
10.6.7.0/24         Direct  0   0         10.6.7.2        GE0/1
10.6.7.2/32         Direct  0   0         127.0.0.1       InLoop0
10.6.7.255/32       Direct  0   0         10.6.7.2        GE0/1
10.6.8.0/24         Direct  0   0         10.6.8.2        GE0/2
10.6.8.2/32         Direct  0   0         127.0.0.1       InLoop0
10.6.8.255/32       Direct  0   0         10.6.8.2        GE0/2
127.0.0.0/8         Direct  0   0         127.0.0.1       InLoop0
127.0.0.1/32        Direct  0   0         127.0.0.1       InLoop0
127.255.255.255/32  Direct  0   0         127.0.0.1       InLoop0
224.0.0.0/4         Direct  0   0         0.0.0.0         NULL0
224.0.0.0/24        Direct  0   0         0.0.0.0         NULL0
255.255.255.255/32  Direct  0   0         127.0.0.1       InLoop0
```

RIP RIP S3 10.6.2.0/24 10.6.3.0/24 10.6.8.0/24 10.6.8.0/24 R1 10.6.5.0/24 10.6.7.0/24 10.6.3.0/24 R2 10.6.2.0/24 10.6.5.0/24 10.6.6.0/24 RIP RIP Cost 1

10.3 RIP



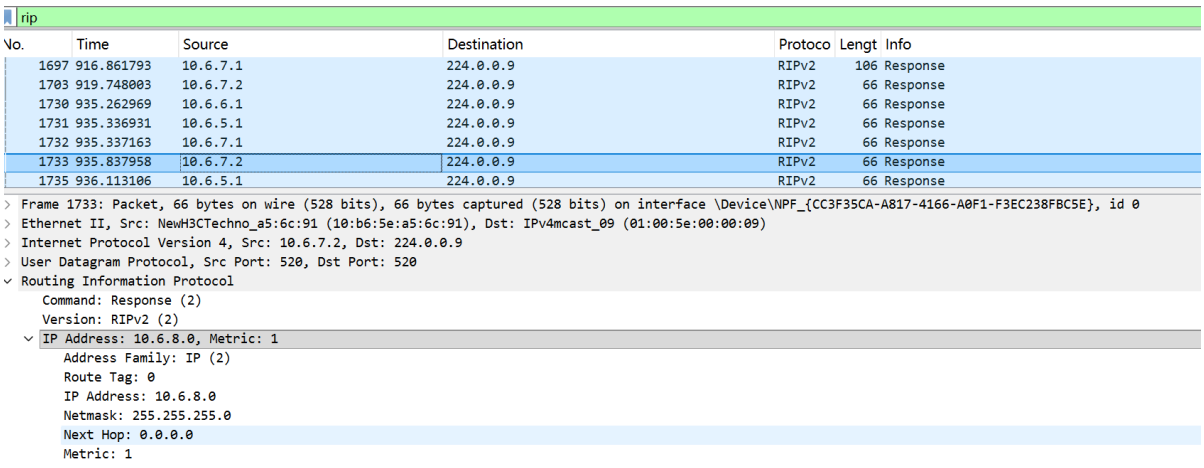




A B 10.6.8.0/24 RIP

10.6.8.0/24 metric

RIP

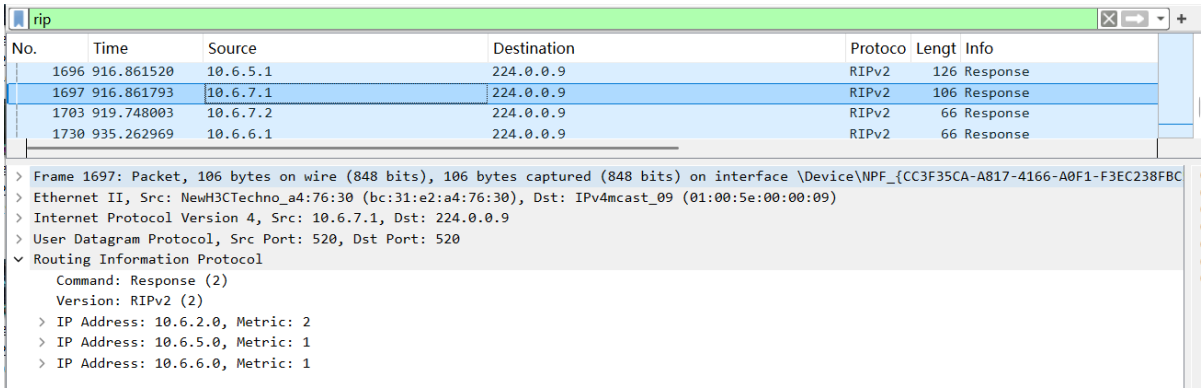
## 10.4 RIP



10.6.8.0/24 10.6.8.0/24 1 2

10.6.8.0/24 RIP Response metric 16
RIP metric=16 “ ” 10.6.8.0/24

10.5 RIP




10.6.8.0/24 RIP
RIP RIP